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ABSTRACT

In an experimental microwave ion source plasma system with a well-defined cavity geometry, multiple cavity resonant mode excitations have been observed. The interactions among these modes can influence microwave coupling to the plasma, enhance plasma uniformity, and affect plasma oscillations. The superposition of closely spaced cavity resonant modes leads to a temporal modulation of the plasma due to the beating effect between pairs of modes. As a result, a new range of plasma oscillations is recorded at the same modulation frequency. This modulation is confirmed by the experimentally measured frequency emission spectra and the accumulation of hot electrons within the plasma-filled cavity. The plasma's resonance with the modulated wave contributes to an increase in the hot electron population. Additionally, the phenomenon of parametric decay (PD) can help explain the rise in hot electron populations in over-dense plasma. The frequency emission spectra show evidence of ion acoustic waves, whose daughter electrostatic waves are resulting from the PD. These appear as two sideband frequency peaks around each excited cavity mode frequency, adhering to the frequency and k-vector selection rules. The observed daughter wave peaks are identified as ion acoustic waves. All experimental findings have been further supported by analytical calculations and microwave plasma simulations conducted using the Finite Element Method in COMSOL Multiphysics software. This paper highlights the temporal phase modulation and the PD phenomena induced by the excitation of different closely spaced cavity modes around the broadly launched microwave frequency of approximately 200 MHz at 2.45 GHz and their interactions.

I. INTRODUCTION

The generation of a high-density and uniform plasma across a large plasma column is crucial in various fields, including high current microwave ion sources, circular wafer processing technologies, surface plasmon resonance sensors, and antenna applications.[1–3] Improved plasma uniformity can be achieved by injecting dual-frequency microwaves and/or modulated microwaves into the plasma cavity. A microwave discharge ion source (MDIS) is required to produce a highly intense, single charge-state, low emittance, and stable ion beam to optimize the performance of accelerators that operate under significant beam loading conditions.[4] To date, several experiments conducted worldwide have demonstrated that plasma non-uniformity and turbulence within the ion source can limit accelerator performance, resulting in reduced ion flux, beam halo formation, parametric instabilities, and beam oscillations, which can lead to transverse emittance growth.[4–6] Achieving plasma uniformity in an MDIS can be accomplished by implementing a microwave discharge cavity that utilizes multiple cavity resonant modes. Recently, experiments involving two close-frequency heating (TCFH) have been conducted[7] to dampen kinetic plasma instabilities and to produce an intense, higher charge state ion beam.

In this study, multiple cavity resonant modes are selectively excited by injecting microwaves (MWs) with a broad frequency range of approximately 200 MHz. The superposition of two closely spaced cavity resonant modes produces a low-frequency wave due to the beating effect. The presence of this low-frequency wave modulates the phase of the resulting cavity mode field temporarily. This low-frequency modulation, also known as temporal phase modulation, causes pulsations in the cavity mode field, which in turn results in plasma oscillations at the same modulation frequency. The presence of phase modulation under over-dense plasma conditions is evidenced by the frequency measured using an RF probe.

The accumulation of hot electrons within the over-dense plasma arises from the resonant interaction between modulated waves and local plasma electrons, as well as the phenomena of parametric decay (PD), which generates electrostatic waves in the plasma. This resonance occurs near the plasma boundary. Notably, the electron cyclotron resonance (ECR) condition corresponding to the launched frequency near the over-dense plasma boundary is not achieved.[7,11] The temporal phase modulation and generation of hot electrons are supported by mathematical derivations and a microwave plasma simulation conducted using COMSOL Multiphysics software, which takes into account similar experimental configurations and operating conditions.[8–10]

In the current microwave ion source plasma system, three distinct excited cavity resonant modes are identified in the frequency emission spectra during plasma operation. By selectively coupling these modes under specific plasma loading conditions, it is possible to control plasma density, uniformity, and distribution throughout the volume [1–2].[11–16] These modes are typically confined near the plasma–sheath interface, at a distance approximately equal to the skin depth in an over-dense plasma state.[17–22] Cortázar *et al.*[11] demonstrated the presence of eight different plasma distribution patterns using an ultrafast imaging technique in a similar microwave discharge cavity. Their findings indicate that the topology and strength of the magnetic field dictate the transitions between different plasma patterns. Additionally, the effects of neutral gas pressure and microwave power are found to be negligible.

This paper presents experimental verification of the temporal phase modulation of cavity resonant modes excited in the over-dense plasma of a microwave discharge ion source (MDIS). These resonant modes are closely spaced around the launched broadband frequency of 2.45 GHz, which has a bandwidth of approximately 200 MHz. The paper is organized as follows: Sec. II describes the theoretical analysis used to interpret the temporal phase modulation, employing analytics and microwave plasma simulations. Sections III and IV discuss the experimental methods and their results, focusing primarily on the phase modulation of multiple cavity resonant modes. Section V explores the physical implications of the experimental results and their significance. Finally, conclusions are presented in Sec. VI.

II. THEORETICAL ANALYSIS

Before verifying the measured frequency emission spectra and illustrating the phase modulation, associated with the cavity modes, a scheme for generating the multiple cavity modes in vacuum cavity (in the absence of the plasma) is described first.

The creation of plasma following microwave injection leads to a change in the electrical permittivity of the cavity system. Importantly, the presence of plasma does not disrupt the resonating characteristics of the cavity-plasma system. Instead, it gives rise to a new set of peaks on the frequency scale. This phenomenon in the plasma-filled cavity is further explained through simulation results.[23–25]

A. Generation of multiple cavity modes within a cavity

Several research groups worldwide are exploring different cavity tuning mechanisms using various methods such as the bias disk technique, varying chamber sizes, and inserting components into the cavity to assess the plasma response and enhance the performance of the ion source. This current study investigates the influence of cavity modes that control field distributions by altering the microwave launcher geometry, the high-voltage break combined with the quartz vacuum window, and the cavity dimensions to achieve high-density plasma (see Fig. 1). In the experimental setup outlined in Sec. III, vacuum continuity is maintained throughout the cylindrical chamber, extending to the quartz vacuum window that interfaces with the ridge waveguide. Broadband microwave radiation (∼200 MHz) at 2.45 GHz is injected from the magnetron side and encounters varying pressure conditions and magnetic field zones after passing through the quartz vacuum window. As the microwaves penetrate the quartz plate, their propagation direction and phase change, leading to the formation of multiple modes within the integrated cavity under pure vacuum conditions (i.e., no gas injection and no plasma). The primary motivation of this work is to understand how multiple vacuum cavity modes influence plasma production and alter its characteristics. Once the plasma is generated, excited cavity modes are observed at different frequency locations. The interactions among these excited cavity modes result in the temporal phase modulation within the microwave plasma cavity. Consequently, this affects plasma stability, which in turn impacts the ion beams produced by the Microwave Discharge Ion Source (MDIS).

The electric field distribution in the presence of multiple cavity modes is simulated to understand the influence of every component of the MDIS (e.g., the cylindrical chamber, Boron Nitride (BN) plate, 4-step ridge waveguide, and an HV break cum vacuum window) that comprises the integrated plasma cavity (Fig. 1).

The microwave simulation is performed using commercial COMSOL Multiphysics software considering the simulation geometry to be similar to the integrated experimental setup.

In Fig. 2, three dominant resonant frequencies are observed near 2.45 GHz within a cavity composed of a cylindrical chamber and a four-step ridge waveguide. Although there are other frequencies where the S_{11} parameter shows dips in the simulation results, their amplitudes are less significant compared to the case at 2.45 GHz. The S_{11} parameter represents the reflection coefficient between the port impedance and the input impedance, as viewed from the source end toward

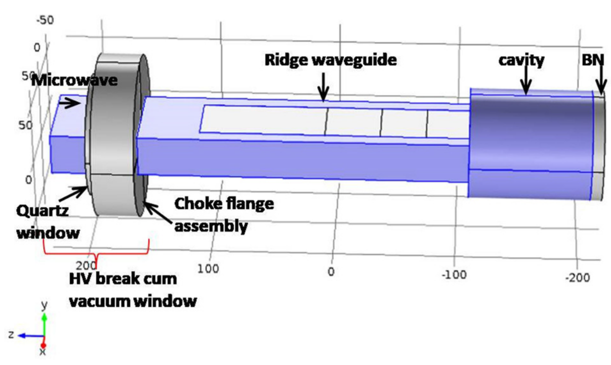

FIG. 1. Computational geometry of the integrated plasma cavity.

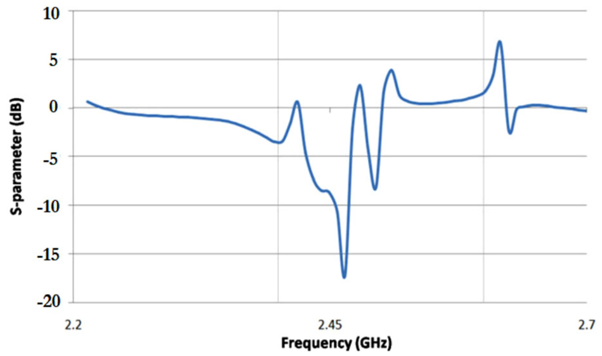

FIG. 2. S_{11} parameter of the integrated cavity comprising of 4-step ridge waveguide and cylindrical cavity.

the load end. In Fig. 3, an additional component—a high-voltage break combined with a vacuum quartz window—is introduced after the four-step ridge waveguide, and the S_{11} parameter is displayed. A quartz circular plate with a diameter of 100 mm and a thickness of 6.3 mm is placed on the waveguide path, located 85 mm from the face of the four-step ridge waveguide. The insertion of the quartz plate into the computational geometry affects both the direction and phase of microwave propagation, resulting in the induction of multiple cavity resonant modes within the integrated cavity under vacuum conditions. These effects are illustrated through the S_{11} parameter plot in Fig. 3.

The S_{11} parameter shows a dip at three frequencies and those are at 2.29, 2.42, and 2.53 GHz frequencies. The simulation study indicates that the integrated cavity (cylindrical chamber + ridge waveguide + quartz vacuum window) can support multiple cavity modes if the launched microwave spectrum is broadband in nature and contains frequencies as identified in the simulation.

The cavity modes of the integrated experimental setup (discussed in Sec. III, Fig. 5) without plasma condition are identified through the microwave simulation that possesses a significant amount of the microwave electric field throughout the cavity volume around the

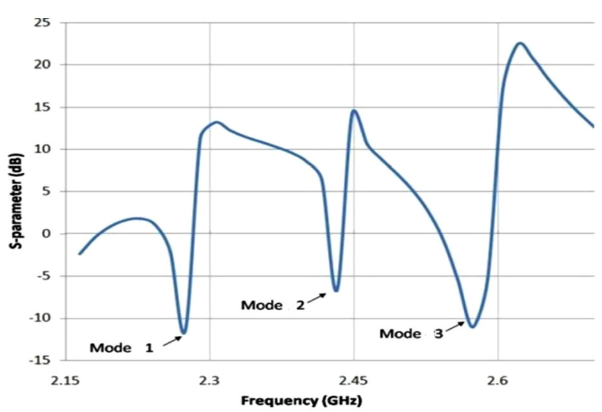

FIG. 3. S_{11} parameter of the integrated cavity comprising of 4-step ridge waveguide, circular quartz plate part of HV break cum vacuum window, BN plate attached to the extraction chamber wall, and a cylindrical cavity. The three modes which can be supported are named as "vacuum cavity modes" (no plasma) and represented as ①, ②, and ③.

launch frequency, 2.45 GHz. The simulation calculates the electric field strength in the r–φ and r–z planes. Out of many cavity resonant modes, which are close to the fundamental frequency, 2.45 GHz (f_0), three close-frequency resonant modes are found to have significant values within the cavity. These modes are of TE_{111} type. From the simulation, it was found that the electric field associated with the three vacuum cavity modes (i.e., mode 1, mode 2, and mode 3) listed in Table I of Ref. 9 has significant intensity. The electric field intensity for the three cavity modes varies in the central region from 55 to 100 kV/m.

B. Excitation of the generated cavity modes by launching broadband microwave frequency

It is in principle not possible to excite other cavity mode frequencies in addition to the single cavity resonant mode frequency with the use of a single microwave launching frequency at 2.45 GHz even within a cavity with well-defined geometry where other dominating cavity resonant modes may be present. The other cavity mode frequencies are possible to be excited only if they were excited somehow by similar pumping frequencies, already existing in the excitation spectrum or self-generated inside the cavity. The excitation of other cavity modes may happen only if the phenomenon is mediated by the plasma, which, by the way, broadens the effective Q-value of the cavity due to the influence of a cavity-plus-plasma system, as a whole.

In an empty multimode cavity, additional cavity modes can be excited because the magnetron spectrum is not monochromatic, particularly at low power levels. The magnetron used in this experimental work has a broad spectral bandwidth of approximately 200 MHz around the fundamental frequency of 2.45 GHz, as illustrated in Fig. 3 of Ref. 26. This extensive spectral range overlaps with the multiple cavity mode frequencies supported in the plasma cavity being studied. In contrast, with a narrow bandwidth (single frequency) magnetron spectrum, excitation of multiple cavity modes is not possible. The additional cavity mode frequencies, such as 2.29 and 2.53 GHz, are excited because the magnetron spectrum includes these frequencies due to its wide bandwidth. When plasma is created, the cavity is unable to support modes at 2.29 and 2.53 GHz. However, three other modes are excited around the launched frequency of 2.45 GHz: 2.4497, 2.4510, and 2.4523 GHz, which are designated as #1, #2, and #3, respectively. These frequencies and their relative intensities compared to the launched frequency of 2.45 GHz are detailed in Table I. The experimental observations related to these modes are discussed in the experimental results section. These excited cavity modes can interact both linearly and nonlinearly, generating various daughter modes. It is

TABLE I. Vacuum cavity (no plasma) and excited cavity (in plasma) modes measured at 400 W and 1.5×10^{-4} mbar.[9]

Vacuum cavity mode frequencies (GHz) and peak No. (no plasma)	Excited cavity mode frequency (GHz) and peak No. (in plasma)	Normalized intensity with respect to f_0 (vacuum and excited mode)
2.295 and ①	2.4497 and #1	0.28 and 1.1
2.42 and ②	2.4510 (f_0) and #2	0.579 and 2.75
2.53 and ③	2.4523 and #3	1 and 2

interesting to predict the outcomes of these interactions theoretically, and attempts have been made to do so. The calculations regarding these predictions are discussed below.

C. Derivation of phase modulation among excited cavity modes

Let two close-frequency modes denoted as #1 and #2 with some phase delay between them superpose. The superposition of two cavity modes results in a modulated wave.[1,2] For example, a mode #1 having an axial B-field (H_z) component at some point on cylindrical axis (r, θ, z) can be written using Bessel's function as[1,2]

$$H_z^{mode\,1} = H_0 J_1(Kr)\cos(\omega_1 t)\cos\theta\cos\beta z. \tag{1}$$

Equation (1) is divided by $H_0 J_1(Kr)\cos\theta\cos\beta z$, which gives its normalized form,

$$B_z^1 = A_1\cos(\omega_1 t). \tag{2}$$

Similarly, the axial B-field normalized of mode #2 having an arbitrary phase delay $\varnothing$ with respect to mode #1 can be written as

$$B_z^2 = A_2\cos(\omega_2 t - \varnothing). \tag{3}$$

The superposition of two modes considering their equal amplitudes ($A_1 = A_2 = A$) gives

$$B = B_z^{\#1} + B_z^{\#2} = \mathrm{A}[\cos(\omega_1 t) + \cos(\omega_2 t - \varnothing)]. \tag{4}$$

By considering the following factors, Eq. (4) can be simplified to Eq. (5) by defining the variables as follows: $\omega_2 - \omega_1 = \omega_{LF}$ where ω_{LF} is much less than both ω_1 and ω_2, and both ω_1 and ω_2 are equal to ω,

$$B = C\cos(\omega_2 t + \psi)]; \tag{5}$$

where $\psi = (\omega_{LF} t + \varnothing)/2; C = \pm 2\cos\varnothing\cos(\omega_{LF} t - \varnothing)/2$.

D. Quantification of phase modulation by simulation

To quantify the phase-modulated wave frequency analytically, a set of Maxwell's equations are simplified to modified Bessel's differential equations in a cylindrical cavity geometry in terms of H_z.[19–21]

The solution to the differential equation at the plasma medium is $H_z = A_1 I_m(Kr)$. Here, $K^2 = (\omega/c)^2(\varepsilon_2^2 - \varepsilon_1^2)\varepsilon_1^{-1}$ and $\kappa^2 = (\omega/c)^2\varepsilon_s$. The terms A_i ($i = 1$ and 2), K^{-1}, $J_m(\kappa r)$ and $N_m(\kappa r)$, $\varepsilon_i (i = 1 \text{ and } 2)$, ε_s, ω, and c are real constants, the penetration depth of MW into plasma, Bessel function of 1st kind, Bessel function of 2nd kind, two components of ($\varepsilon_1 = 1 - \Omega_\alpha^2/(\omega^2 - \omega_\alpha^2)$, $\varepsilon_2 = -\Omega_\alpha^2\omega_\alpha/\omega(\omega^2 - \omega_\alpha^2)$) permittivity present in a microwave plasma cavity,[19–21] the permittivity of the sheath, wave frequency, and speed of light, respectively. Here, Ω_α and ω_α are the plasma and cyclotron frequencies of plasma particle species α ($\alpha = i$ for ion and e for electrons). The condition $K^2 > 0$ determines the possible frequency ranges at which the wave propagates at the interface region of the plasma boundary. The solution to this inequality gives two frequency ranges, which are $\omega_{LF} < \omega < |\omega_e|$ and $|\omega_e| < \omega < \omega_1 - |\omega_e|$, respectively. Here, *LF* stands for low frequency and $\omega_{LF} (= 2\pi f_{LF})$ falls in the lower hybrid (*LH*) frequency range ω_{LH} and it can be estimated as $\omega_{LH} = [\Omega_i^{-2} + (\omega_i|\omega_e|)^{-1}]^{-1/2}$ and $\omega_1 = 0.5\,\omega_e + [\Omega_e^2 + 0.25\,\omega_e^2]^{1/2}$.

For some typical simulated plasma parameters, i.e., see Fig. 4(a), $n_i = 5.5 \times 10^{15} m^{-3}$, $B = 0.23T$ near the plasma boundary wall, a typical modulated wave frequency is estimated which can be supported for propagation near the plasma sheath. The estimated modulated wave frequency is $\omega_{LF} = \sim 1.3$ MHz. Microwave plasma modeling description is provided in Refs. 8 and 9. Simulation is performed considering similar experimental configuration and operating parameters using COMSOL Multiphysics software©. Then, Eq. (5) shows the phase of the resultant axial B-field ψ is modulated with frequency, ω_{LF}. Similarly, the phase of the resultant electric field (E) is also modulated at ω_{LF}.[21,22] Hence, the force exerted on the electrons by the resultant electric field of the modulated wave where plasma resonance conditions, i.e., $\omega \gg \omega_{pe}$ and $\omega_{pe} = \omega_{LF}$,[21,22] are satisfied is given by the following function:

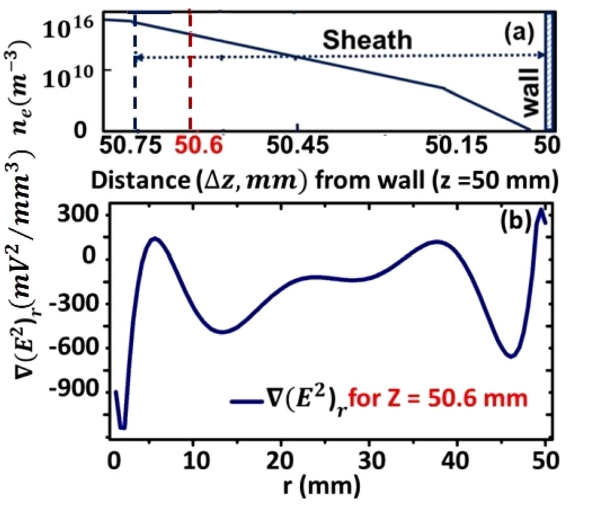

FIG. 4. (a) Schematic of plasma sheath near wall ($z = 50$ mm). (b) Radial gradient ($\nabla(E^2)_r$) of the strongly inhomogeneous electric field within the plasma sheath.

$$f_{avg}(t) = \omega_{LF} * \left(e^2/2m_e\omega^2\right)\nabla\left(E^2(r - \omega_{LF} t)\right). \tag{6}$$

This force is calculated by using simulated results shown in Fig. 4. Figure 4(b) gives a radial variation of $\nabla(E^2)_r$ within the plasma sheath thickness [Fig. 4(a)].

The force using Eq. (6) is estimated as $49 \times 10^{-23} * \nabla(E^2)_r\,N$ for some typical estimated parameters, such as $f_{LF} = 1.3$ MHz and $\nabla(E^2)_r = 300 \times 10^3 V^2/m^3$ (Fig. 4). The electron acceleration via the plasma resonance due to this force raises the electron temperature to a high value near the plasma boundary.[27] This force is exerted on the electrons with a distance of plasma column dimension (say 50 mm for example as shown in Fig. 4(b). So, the average energy (force × distance) gained by electrons is equal to $\frac{3}{2}k_B T_e[K]$. Finally, the electron temperature value for the above parameters is estimated to be 32 eV near the plasma boundary due to modulated wave resonance with the local plasma electrons.[28] The experimental pieces of evidence of the modulated wave and the signatures of the high-energy electron production are presented in Sec. IV.

III. EXPERIMENTAL METHOD

A schematic view of MDIS and its plasma diagnostic setup is shown in Fig. 5.[8] Figure 5 shows four ring magnets (1.38 T each, magnetization is along the thickness) surrounding the plasma chamber coaxially in mirror-B configuration mode. In this experiment, right-hand

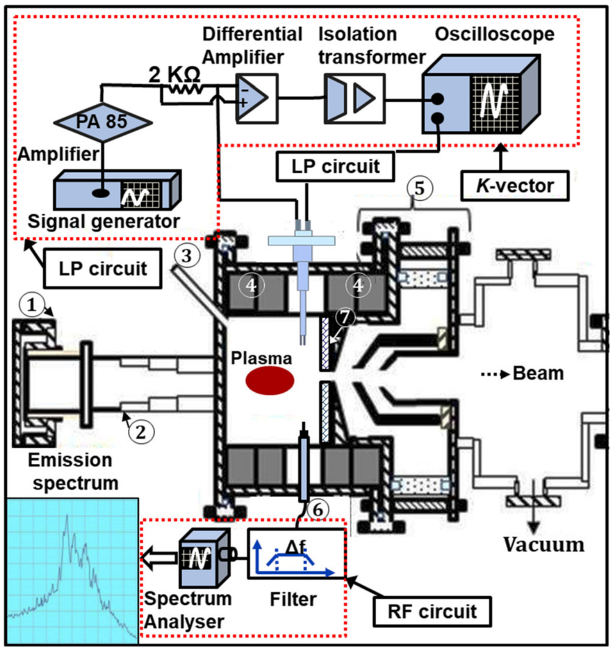

FIG. 5. Schematic of plasma diagnostics setup of MDIS. (1). HV break and vacuum window; (2) four-step ridge guide; (3) gas inlet; (4) ring magnet pairs; (5) ion extraction grids; (6) RF probe; (7) boron nitride (BN) plate. The setup is also equipped with a double Langmuir probe (LP) to measure the k-vector of the excited modes.

polarized MW (*R*-wave) of the frequency 2.45 GHz is launched using the microwave launcher to the cavity from the high B-field side of one set of the permanent ring magnet.[8,9]

The launcher geometry includes the four-step ridge waveguide and the high voltage (DC) break cum vacuum window that connects the cavity. For ion beam extraction purposes, the launcher unit and the cavity are integrated with a vacuum chamber through an aperture having a dimension of a few millimeters. The vacuum chamber consists of a three-grid ion extraction system and the beam emittance diagnostic setup. A boron nitride (BN) plate is placed at the inner plasma-grid surface to get a higher extracted ion current density.[16] The description of the setup is shown in Fig. 5. However, it is to be noted that the present experimental setup is working on the ion source scheme without the use of the sextuple/octupole magnetic field configuration, normally used in ECR ion sources.

If the launched MW is exactly parallel to the B-field axis, it can propagate toward the dense plasma region. However, due to the mirror field configuration in the present setup, the MW axis does not exactly coincide with the magnetic axis. As a result, both the extraordinary (*X*) mode and the ordinary (*O*) mode would exist in the plasma irrespective of launching a pure *R*-wave. The breakdown of nitrogen gas in a vacuum is done by injecting MW power (set power) from 50 to 700 W. The power reflection from plasma ranges between 5% and 10% with the increase in set power.[8] A more detailed description of the experimental setup and the diagnostic methods can be found in Ref. 8.

During the experimental campaign, the frequency emission spectra from plasma are measured by a probe tip (also denoted as RF probe). The probe tip is placed near the beam extraction grid location (r, $z = 0$, 25 mm), as shown in Fig. 5. The probe detects frequency emission from the cavity with and without plasma. The frequency signal collected by the probe is analyzed using an RF circuitry consisting of a microwave spectrum analyzer (SA) (model: FSH8, make: ROHDE & SCHWARZ, band: 100 Hz-8 GHz) and FSH4View data acquisition software. A bandpass filter (LORCH MICROWAVE make, model: 9IZ3-2500/A1000-S) is employed to select a particular frequency band around 2.45 GHz. Despite launching 2.45 GHz MW ($\omega_0 = 2\pi f_0$) (bandwidth ∼10–25 MHz[8]), different cavity resonant modes with and without plasma are observed inside the cavity around 2.45 GHz.

The chamber is maintained at base pressure $\sim 1 \times 10^{-7}$ mbar and the observed frequency emission peaks reported here are in the operating pressure range of 2×10^{-4}–1×10^{-3} mbar.[8] The floating potentials of plasma are recorded by a two-tip-based Langmuir probe. The LP has two tips. Two tips separated by ∼1 mm are used as two independent Langmuir probes as shown in Fig. 5. Time-varying floating potential signals are separately coming from the two probe tips simultaneously and acquired by an oscilloscope. The phase difference of the floating potential of each probe divided by their separation measures the wavenumber k.[8] The k-vector of a plasma wave is constructed by aligning the plane of the two probes in the perpendicular (denoted as k) as well as parallel (symbolized by $k_{\|}$) direction with respect to the B-field.

IV. RESULTS

As discussed before, the frequency emission from the plasma is recorded by an RF probe for different set powers. In the beginning, a spectrum in vacuum condition is obtained to record the presence of different cavity resonant modes (also called vacuum cavity modes (VCMs)) inside the cavity,[9] to understand the basic characteristics of the integrated geometry. The VCMs are given in Table I.[9] With increase in the set power (100–600 W), the plasma moves from underdense condition to over-dense condition. During the plasma discharge, modified characteristics of the frequency spectrum emitted from the plasma are shown in Fig. 6 and also noted in Table I for 400 W magnetron set power.

The plasma buildup after the microwave injection causes a change in the electrical permittivity of the cavity system with a well-defined geometry. Experimental data say the plasma creation does not destroy the resonating structure of the cavity-plasma system. Some vacuum resonant modes disappear and a few cavity modes are excited in the plasma. Hence, the power coupling changes between the magnetron source and the plasma. The variation in peak heights and widths of the excited cavity modes, observed in the frequency emission structures, reflects these changes in power coupling. Specifically, the intensity peak height can change by as much as 10 dB, and the width can vary by several hundred kilohertz, depending on the set power and operating pressure. Furthermore, the electromagnetic frequency emission spectrum can be influenced by the distribution pattern of the plasma density.

It is important to note that RF probe measurements taken from a fixed position within the plasma may provide incomplete information about the plasma parameters. The inhomogeneous plasma profile and the magnetic field play significant roles in determining the frequency emission structure, as they also affect the propagation and coupling of electromagnetic waves to the plasma. As shown in Fig. 6, the measured emission spectrum reveals a frequency spacing of approximately

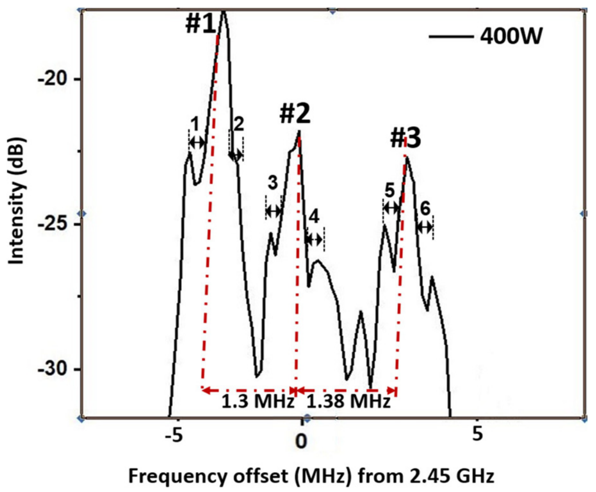

FIG. 6. Frequency emission spectrum around 2.45 GHz (f_0) shows the three cavity mode peaks (named as #1, #2, and #3) in the plasma and their respective upper and lower sideband range denoted as 1, 2, 3, 4, 5, and 6); two side peakss are around every excited cavity mode peak. The spectrum is recorded at 6×10^{-5} mbar pressure, 400W, $\sim$0.06T B-field, and probe position (r, z) = (10 and 25 mm). Resolution bandwidth (RBW) and its span are kept at 10 kHz and 10 MHz, respectively.

1.3 MHz between consecutive excited cavity modes in the presence of plasma.

The frequency spacing aligns with our prediction of the phase modulation frequency discussed in Subsection II C, which supports the idea of linear interactions among the excited cavity modes. The phase-modulated wave, observed in the frequency emission signal obtained by the RF probe, has a frequency of approximately 1.3 MHz, as shown in Fig. 6.[1,2] As illustrated in Fig. 6 and Table I, three distinct cavity mode peaks can be identified: Peak #1 at 2.4497 GHz, Peak #2 at 2.4510 GHz, and Peak #3 at 2.4523 GHz. The frequency spacing between each consecutive cavity mode peak—specifically between Peak #1 and Peak #2, as well as between Peak #2 and Peak #3—is approximately 1.3 MHz.

Additionally, it is observed that every excited cavity mode peak has both an upper and a lower sideband peak. These sideband peaks are represented by a frequency difference from the parent excited cavity mode, denoted as follows: 1 (555 kHz), 2 (238 kHz), 3 (476 kHz), 4 (317 kHz), 5 (397 kHz), and 6 (873 kHz), as shown in Fig. 6. The frequencies of the sideband peaks arise from the interaction between waves with frequencies in the range of a few hundred kilohertz and the parent excited cavity modes. The low-frequency waves, as shown in Fig. 6, are identified as ion acoustic waves (IAWs) based on the dispersion relation discussed in Refs. 8, 9, and 31. Furthermore, the sideband frequencies associated with IAWs are confirmed through the relationship between electron temperature and the ion acoustic speed, as explained in Ref. 9.

The observed sideband peaks can be explained by the phenomenon of parametric decay (PD). During PD, the excited cavity mode wave or the launched wave can generate two electrostatic (ES) modes: one associated with electrons, identified as the electron Bernstein wave (EBW), and the other associated with ions, referred to as the ion acoustic wave (IAW). In the current experimental setup, the IAW signal is detected by an RF probe analyzing the collected emission spectra. Additionally, a double Langmuir probe is used to identify the IAW by measuring the k-vector within the plasma. Unfortunately, there are insufficient diagnostics available in the current setup to identify the EBW.

The conversion from electromagnetic (EM) mode to electrostatic (ES) mode takes place on the surface of the upper hybrid resonance (UHR) zone ($n_e < n_{critical}$ and $B < B_{ECR}$).[9–12] The critical density layer ($n_{critical}$) corresponds to the local plasma frequency that becomes equal to the launched frequency (2.45 GHz) in an inhomogeneous plasma volume. The critical density layer is treated as the cutoff layer of EM wave propagation if the frequency of the EM wave is equal to or less than the plasma frequency. For 2.45 GHz microwave frequency $n_{critical} = 7.4\times10^{16}$ m^{-3}. The magnetic field B = 875 G is the B_{ECR} boundary inside the cavity volume where the electron cyclotron resonance (ECR) condition is satisfied. When the launched 2.45 GHz MW wave is reflected from the cutoff layer, it reaches near the UHR zone and EM to ES mode conversion takes place.[12,29]

If f_{IAW} is the measured IAW frequency generated due to PD, it may interact with the parent excited cavity mode f_0 and produce two daughter waves of frequency $f_2 = (f_0 - f_{IAW})$ and $f_3 = (f_0 + f_{IAW})$. These two daughter waves (f_2, f_3) can appear in the emission spectra as two sideband peaks. The PD process follows also the k-vector selection rules, i.e., $k_2 = (k_0 - k_{IAW})$, and $k_3 = (k_0 + k_{IAW})$. From the experimental data, it is found that the measured k-vectors also follow the selection rules.[12]

Since the two consecutive cavity modes can interact with each other, it can effectively create the temporally phase-modulated wave with the frequency of $\sim$1.3 MHz.[1,2] To support this, a frequency emission signal is acquired by a Langmuir probe and a spectrum analyzer circuit assembly (Fig. 5). The emission signal is shown in Fig. 7(a). The intensity of that modulated wave increases with the set power. The probe is also used to generate the electron energy distribution functions (EEDFs) for different powers and is shown in Fig. 7(b). A relative variation of the bulk plasma temperature and the hot electron temperature for different set power is shown in Fig. 7(c).

It has been observed in Fig. 7Ⓒ that the population of hot electrons decreases when the magnetron set power is increased from 100 to 250 W. However, with further increases in power, the temperature of the hot electrons rises, stabilizing within the range of 25–35 eV. In contrast, the bulk plasma electron temperature shows a consistent increase, rising from approximately 1 eV to about 5 eV as the power is increased from 100 to 600 W. Phase modulation is considered one possible explanation for the generation of hot electrons.[21,22,27] Additionally, parametric decay may also contribute to the presence of high-energy electrons and the related study is reserved for future scope of work.

V. DISCUSSION

As presented above, the phase-modulated wave with a frequency of approximately 1.3 MHz results from the superposition of pairs of closely spaced excited cavity modes present in the plasma (see Fig. 6). This wave propagates near the plasma boundary regions and has been experimentally observed to influence plasma dynamics and temperature as it travels through these areas. The reason for this influence is that the phase-modulated wave couples with the local plasma electrons, transferring energy to them via plasma resonance near the boundary, where the density is lower and contributes to the presence

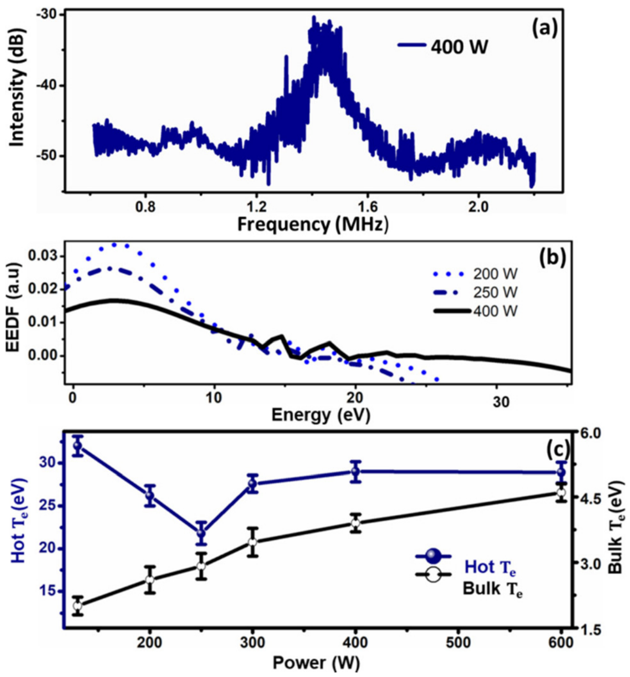

FIG. 7. (a) Spectrum acquired by the spectrum analyzer, (b) electron energy distribution function (EEDF) obtained from Langmuir probe data for different set power, and (c) variation of bulk and hot electron temperature (T_e) with set power.

of hot electrons (refer to Fig. 7). Consequently, this interaction can generate a temporally modulated inhomogeneous electric field near the boundary. The radial gradient of the electric field, illustrated in Fig. 4(b) as part of the simulation findings, supports this explanation.

The observations of the ion acoustic wave (IAW), which is responsible for the two sideband peaks observed around each excited cavity mode (see Fig. 6), can be explained using existing literature on parametric decay[15,32,33] as well as electromagnetic field simulations conducted using COMSOL Multiphysics software. ALIEV and SLLIN'S theory[15,32,33] described that the ion acoustic wave frequency range can exist if an intense MW $\tilde{E}$-field (i.e., $x \geq \lambda_{De}$) is present in the plasma and k-vector has a large value ($k_{IAW} \approx \lambda_{De}{}^{-1}$). Here, $x\,(=e\tilde{E}/m_e\omega_0{}^2)$ and $\lambda_{De}(\upsilon_{Te}/2\pi f_{pe})$ are amplitude of the electron oscillation under electromagnetic ($\tilde{E}$) field and the electron Debye length, respectively. Here, microwave plasma simulation shows the $\tilde{E}$-field to vary from 120 kV/m to 20 kV/m with increase in the density from $\sim 1\times 10^{16}$ to $1.3\times 10^{17} m^{-3}$ for 50–150 W set power respectively.[8,35–37] Hence, the amplitude (x) of the electron's oscillation can vary from ~90 to 37 μm for the above range of the $\tilde{E}$-field. On the other hand, the measured data show λ_{De} varying from 50 to 36 μm for a fixed thermal electron temperature of 3.5 eV with a density transition from $7.3\times 10^{16} m^{-3}$ (130 W) to $1.5\times 10^{17} m^{-3}$ (180 W), respectively. It implies the $\tilde{E}$-field is intense inside the cavity as $x \geq \lambda_{De}$ and hence the consideration of the sidebands as ion wave is justified for the aforesaid conditions. The frequency f_{ion} depends on x for a given λ_{De} as per the relation $f_{IAW}{}^2 = f_{pi}{}^2\,\left(1-\frac{J_0^2(k_{IAW} * x)}{1+(k_{IAW} * \lambda_{De}{}^2)}\right)$.[15,32] Moisan and others[15,33–37] also concluded that IAW can reach f_{pi} (ion plasma frequency) even

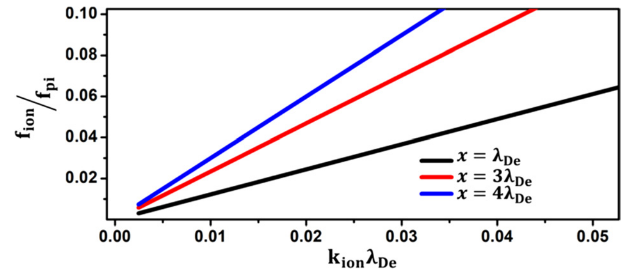

FIG. 8. Ion wave dispersion under intense MW $\tilde{E}$-field ($x \geq \lambda_{De}$).

within the region, $k_{IAW}\lambda_{De} \leq 1$. Since, $k_{IAW}\lambda_{De} \ll 1$, the IAW is unable to reach f_{pi} range in the over-dense plasma (see Fig. 8) for the set power >180 W. Hence, the IAWs represent the sideband values as shown in Fig. 6.

The phenomena of temporal phase modulation and parametric decay (PD) arise from the interaction among three excited cavity modes in a highly inhomogeneous plasma. Both linear (phase modulation) and nonlinear (parametric decay) interactions between pairs of these excited modes can significantly affect the overall dynamics of the plasma and the power coupling processes. Electrostatic (ES) waves can couple power to the plasma through PD, even exceeding the cutoff density limit for microwave penetration, thereby facilitating the creation of over-dense plasma.

From the literature survey, it is understood that the parametric decay phenomenon is associated with the generation of the electron Bernstein waves (EBWs) along with the ion waves.[12,16,29] The EBWs are often depicted as the cause of hot electron generation in the magnetized plasmas.[12,29–33] So, there is a possibility that the electron Bernstein wave is generated from the parametric decay of cavity modes in the present plasma also. However, the existing Langmuir probes are not able to detect the EBWs in the over-dense plasma. This is because the EBW wavelength is much shorter than the presently used Langmuir probe tip length (4 mm).[1] It is known from the literature survey that parametric decay can also generate other types of ion waves, i.e., ion Bernstein waves (IBWs) or Lower Hybrid oscillations (LHOs).[29–37] However, studies on the presence of these waves in the present experimental setup are reserved for future work.

VI. CONCLUSION

Temporal phase modulation of closely spaced cavity modes has been observed in a microwave plasma experiment featuring a well-defined cavity. An experimental modulation frequency of approximately 1.3 MHz has been recorded, which is supported theoretically by the principle of two-wave superposition for close-frequency cavity modes. The resonance of plasma electrons with the modulated wave near the plasma boundary leads to an increase in hot electrons in that region. The generation of these hot electrons has been confirmed by an RF probe positioned near the boundary of the plasma source.

To estimate the hot electron temperature analytically, the electromagnetic field simulation with and without the plasma is performed, which shows the strongly inhomogeneous electric field near the plasma boundary. The simulation is performed by COMSOL Multiphysics software assuming computational parameters similar to the experimental configuration and operating conditions. The temperature of the hot electrons is estimated analytically and is found to vary within the range of tens of eV. It is concluded that the plasma resonance with the modulated wave near the boundary creates a strong inhomogeneous electric field in the sheath and is responsible for the hot electron production. The $\nabla^2 E$ force generated by the electric field due to phase modulation can generate high-temperature electrons (T_e up to 36 eV) that reach maximum value near the plasma boundary. Consequently, one should expect the hot electrons to be generated at the plasma boundary only. However, from several experimental campaigns, it is found that those hot electrons are present across the plasma volume and not only at the boundary. Since the magnetic confinement is poor in the present setup, it is hard to trust that the electrons generated near the walls could fill the whole plasma volume after a back-and-forth reflection. So, it seems that the phase modulation cannot only explain all the hot electron generation in the plasma but just a portion of it. Hence, other possible phenomena like parametric decay may be responsible for explaining the hot fraction of electrons present in the microwave plasma system with well-defined cavity geometry.

The frequency emission from the plasma reveals fine structures of sideband peaks, associated with the cavity modes. This emission provides valuable information regarding the absorption of microwave power by the plasma. It has been observed that the three cavity modes in the highly inhomogeneous plasma display fine structures in the frequency emission spectrum, indicating the presence of electrostatic ion acoustic waves (IAWs) generated through the parametric decay process within the plasma volume.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

C. Mallick: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **M. Bandyopadhyay:** Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **R. Kumar:** Funding acquisition (equal); Project administration (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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